Video Transcript

Video 7: Bayes' Theorem Solution (02:51)

So, the Harvard MBA students said that only 10% of happy people are rich. And, the MIT student said I need more information, and googled and found out that 40% of all people are happy and 5% of all people are rich. So, what did the student actually do then? Well, the student is going to take advantage of Bayes and Bayes says that if we have a prior probability, so probability of something B and now we see that say A has occurred. And we ask now, what is the probability of B, given that A has already occurred? Our answer is probability of B given A, and the new information now gives us a new probability of B.

So, given that A has occurred, probability of A, and the probability of B given A, gives us a new probability for A given B. And this is how we write it. So, let's take a look at the Harvard student and the MIT student. What we really are asking is given that we are rich, what's the probability that we are happy? So, let's call rich R and happy H. We want to calculate P H given R.

So, here we use Bayes, and that's the same as P R given H, probability that we're rich, given that we're happy, probability that we're happy divided by the probability that we're rich. Only 10% of happy people are rich. So, given that we're happy, the probability that we're rich is 10%, 0.1. The probability that we're happy is 0.4. The probability that we're rich, just of all the people, just the normal of anyone in the population, is 0.05. Now when you divide this out, the answer is 0.8. So, 80% of rich people are happy, according to this data.

Now, I'm sure we'll have volunteers that would test this out for us. We'll make them rich and we'll see if they're happy. Okay, so that's the answer to how many rich people are happy. What's the percentage? 80% seems like they're a happy bunch. Okay, so hope you enjoyed the exercise. Bye for now.

© MIT xPRO® 2024. All rights reserved.